

Original Research Article

Q1 Vitamin D₃ and marine ω-3 fatty acids supplementation and leukocyte telomere length: 4-year findings from the VITamin D and Omega-3 Trial Q8 (VITAL) randomized controlled trial

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A B S T R A C T

Background: Limited studies suggest that vitamin D or omega 3 fatty acids (n-3 FAs) supplementation may be beneficial for telomere maintenance; however, evidence from large randomized clinical trials is lacking. We hypothesized that vitamin D or n-3 FAs supplementation reduce leukocyte telomere length (LTL) attrition over time by leveraging the VITamin D and Omega-3 Trial (VITAL) trial.

Methods: VITAL is a large, randomized, double-blind, placebo-controlled trial with a 2 × 2 factorial design of vitamin D₃ (2000 IU/d) and marine n-3 FAs (1 g/d) supplements for 5 y among a representative sample of 25,871 US females aged ≥55 and males aged ≥50 y. The VITAL Telomere study (NCT04386577) included 1054 participants who were evaluated in person at the Harvard Clinical and Translational Science Center. LTL was determined by the Absolute Human Telomere Length Quantification Quantitative Polymerase Chain Reaction method at baseline, year 2, and year 4. The prespecified primary outcome measures were changes in LTL between baseline, year 2, and year 4. Analyses of intervention effect used mixed-effects linear regression models.

Results: LTL was measured in a total of 2571 samples from the 1031 participants at baseline, year 2, and year 4. Compared with placebo, vitamin D₃ supplementation significantly decreased LTL attrition by 0.14 kilo base pairs (kb) (0.01, 0.27) over 4 y ($P = 0.039$). Overall trend analysis showed that the vitamin D₃ supplementation group had LTLs that were ~0.035 kb higher per year of follow-up compared with placebo group (0.002, 0.07, $P = 0.037$). Marine n-3 FAs supplementation had no significant effect on LTL at either year 2 or year 4.

Conclusion: Four-y of supplementation with 2000 IU/d vitamin D₃ reduced telomere attrition by 140 bp, suggesting that vitamin D₃ daily supplementation with or without n-3 FAs might have a role in counteracting telomere erosion or cell senescence.

This trial was registered at clinicaltrials.gov as NCT04386577.

Keywords: vitamin D, marine n-3 fatty acids, telomere length, aging

Introduction

Telomeres, specialized chromatin structures located at the chromosomal ends, protect chromosome integrity and stability. Telomeres naturally shorten with every cell cycle, and cells with critically short telomeres undergo replicative senescence and apoptosis [1].

Telomere shortening has been proposed as a mechanism for decreasing chromosomal stability, which increases risks for chronic diseases, cancer, cardiovascular disease, and overall mortality [1,2].

Leukocyte telomere length (LTL) represents a key integrating component of genetic factors and the cumulative effects of environmental, lifestyle, and nutritional factors throughout human life. Therefore, it is crucial to identify the factors that can slow down telomere shortening, which in turn could prevent premature aging or age-related diseases.

Both vitamin D and omega-3 fatty acids [n-3 (ω-3) FAs] are important for a range of vital cellular processes, including cellular differentiation, proliferation, and apoptosis, and have been postulated to be protective for aging and age-related chronic diseases.

Abbreviations: CTSC, Clinical and Translational Science Center; DBP, diastolic blood pressure; 25(OH)D, 25-hydroxyvitamin D; LTL, Leukocyte telomere length; n-3 FAs, marine omega 3 fatty acids; SBP, systolic blood pressure; VITAL, VITamin D and Omega-3 Trial.

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Associations between telomere length and vitamin D concentrations [3–5] or n–3 FAs [6–9] have been reported in cross-sectional studies. A few short-term small-scale intervention studies with 8 wk to 12 mo supplementation of vitamin D [10–12] and n–3 FAs [13] showed beneficial effects on telomere length or telomerase activity. However, a recent review concluded that the results of the studies performed to date are inconsistent [14]. Moreover, large randomized controlled trials with longer treatment duration remain scarce.

The VITamin D and Omega-3 TriaL (VITAL) is a completed large, randomized, double-blind, placebo-controlled trial with a 2 × 2 factorial design of vitamin D₃ (2000 IU/d) and marine n–3 FAs (1 g/d) supplements for 5 y among a representative sample of 25,871 US females aged ≥55 and males aged ≥50 y. Autoimmune diseases, characterized by an inflammatory autoimmune response to self-tissues, and cancer are 2 chronic diseases that increase in prevalence with age. In the VITAL trial, compared with placebo, daily supplementation with 2000 IU/d vitamin D, but not n–3 FAs, reduced the incidence of

advanced (metastatic or fatal) cancer by 17% [15]. Moreover, supplementation with vitamin D with or without marine n–3 FAs for 5 y reduced all incident autoimmune diseases by 22%, whereas n–3 FAs supplementation with or without vitamin D reduced the autoimmune disease rate by 15% (not statistically significant) [16]. Furthermore, our VITAL ancillary study (NCT04386577) found that supplementation of vitamin D₃ with or without n–3 FA decreased circulating high sensitivity-C-reactive protein concentrations by 19% at year 2 [17]. As such, we tested whether long-term vitamin D₃ or n–3 FAs supplementation reduced LTL attrition in VITAL.

Methods

Study population and design

The VITAL study design, baseline characteristics, and main results have been published previously [18–21]. In brief, VITAL was a

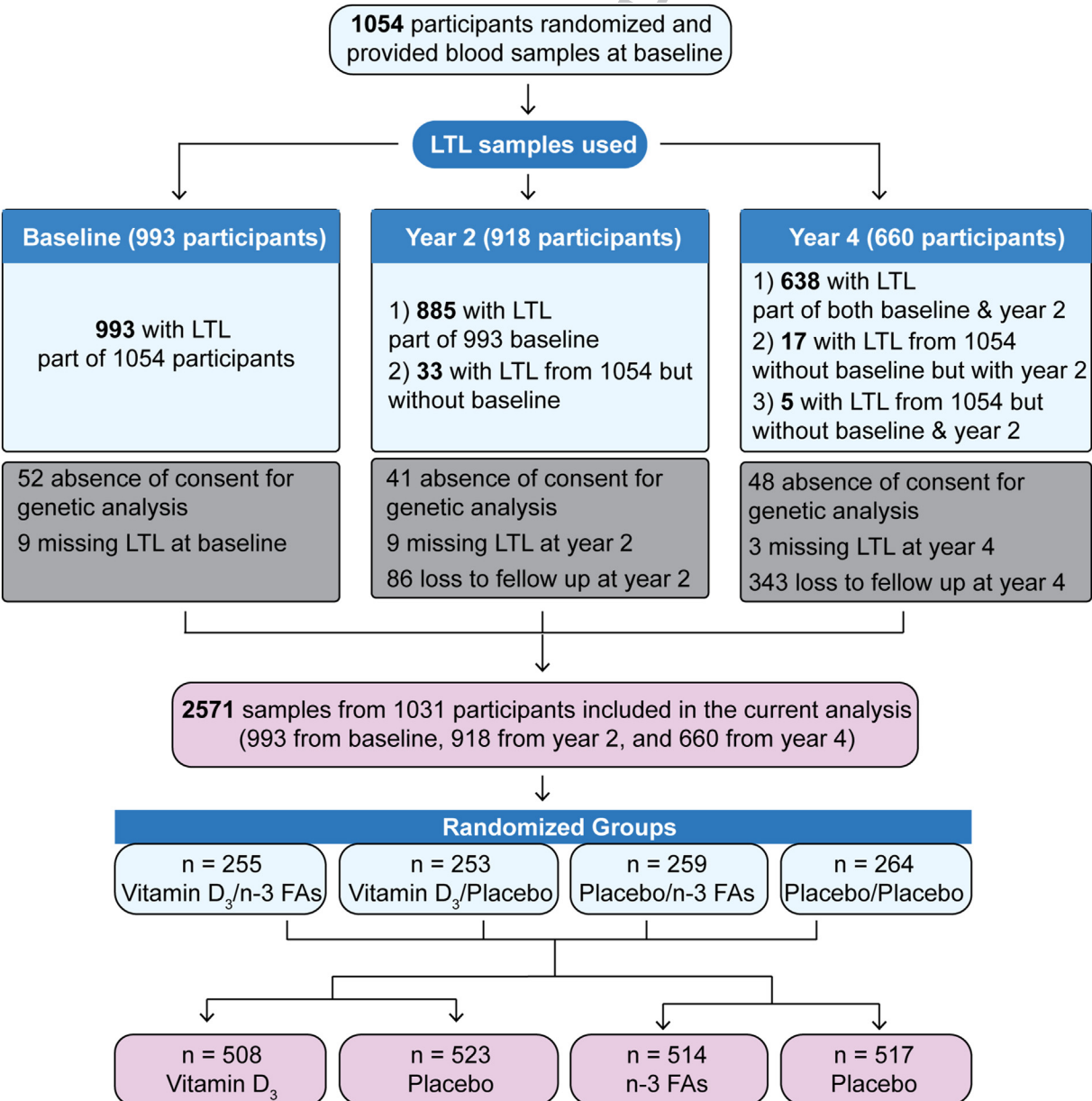


FIGURE 1. Flow diagram of VITAL-CTSC participants. LTL, Leukocyte telomere length.

completed nationwide, randomized, double-blind, placebo-controlled trial of the benefits and risks of supplemental vitamin D₃ (2000 IU/d) and marine n-3 FAs (1 g/d Omacor fish-oil capsule with 840 mg n-3 FAs, including EPA, 460 mg + DHA, 380 mg) in the primary prevention of cancer and cardiovascular disease among 25,871 US males and females, aged ≥ 50 and ≥ 55 y, respectively (NCT01169259) [18, 19]. Eligible participants had no history of cancer (except non-melanoma skin cancer) or cardiovascular disease at trial entry, and they were required to agree to limit the use of vitamin D from all supplemental sources, including multivitamins, to 800 IU/d and to complete a 3-mo placebo run-in phase. Safety exclusions included renal failure or dialysis, cirrhosis, history of hypercalcemia, or other serious conditions that would preclude participation. Participants were randomly assigned to vitamin D₃, n-3 FAs, both active agents, or both placebos in a 2 × 2 factorial design with a 1:1:1:1 allocation ratio. Randomization was performed by the central coordinating center at Brigham and Women's Hospital, and participants were enrolled by study staff at Brigham and Women's Hospital. Randomization took place from November 2011 to March 2014 and was computer generated within sex, race, and 5-y age groups in blocks of 8, with 2 individuals in each of the 4 treatment combinations [20]. Everyone involved in the trial was blinded to interventions throughout the trial (participants, investigators, endpoint adjudicators, and care providers). Intervention ended as planned on 31 December 2017, yielding a median treatment period of 5.3 y (range, 3.8–6.1 y). During the trial, yearly questionnaire response rates averaged 93%, and mortality follow-up rates exceeded 98%. The percentage of participants who reported taking at least two thirds of their study capsules averaged 80% during the intervention period, and <5% and 10% of participants reported outside vitamin D₃ use at 2 and 5 y, respectively [19]. This ancillary study (NCT04386577) aimed to investigate the effect of vitamin D and n-3 FAs supplementation on telomere aging in the Harvard Clinical and Translational Science Centre (CTSC) participants. The sample size was predetermined to include all 1054 participants of VITAL-CTSC cohort at baseline and all those returning and having blood collections at 2 and 4 y. The CTSC visits provide a valuable opportunity for detailed inperson assessments

and blood collections at baseline, year 2, and year 4 follow-up visits. The CTSC subcohort was randomized separately into the 4 treatment groups created by the factorial design (259–268 participants per treatment group). Similar to the randomization procedure in the overall cohort, the CTSC randomization was computer generated within sex, race (African American compared with not), and age group (age <65 compared with ≥ 65 y) in blocks of 8 [21]. The Partners' Health Care System's Institutional Review Board approved this ancillary study. All participants provided informed consent.

Blood collection and laboratory analysis

Fasting blood samples were immediately processed and stored at -80°C at the CTSC. LTL was measured using the ScienCell's Absolute Human Telomere Length Quantification qPCR Assay Kit (ScienCell Research Laboratory, Figure 1). Absolute Human Telomere Length Quantification is designed to directly measure the average telomere length of a human cell population. The telomere primer set recognizes and amplifies telomere sequences. The single copy reference primer set recognizes and amplifies a 100 bp-long region on human chromosome 17 and serves as reference for data normalization. The reference genomic DNA sample with known telomere length serves as a reference for calculating the telomere length of target samples [22]. The lower limit of quantification is around 2 kilobase pairs (kb), and the upper limit is around 20 kb. For quality control, the baseline, year 2 and year 4 leukocyte DNA samples from the same participant were measured in the same plate in duplicates. If the standard deviation of Ct values was >0.5 , the samples from the same participants at 3 timepoints were repeated at the same time. All experiments were performed in triplicates under the same conditions.

Statistical analysis

The current study followed a modified intention-to-treat principle that included 1031 participants with LTL data from at least one timepoint in the analysis. The general characteristics of the participants are presented as mean $\pm$ SD for continuous variables and *N* (%) for categorical variables. Baseline differences of general characteristics

TABLE 1

Characteristics of the CTSC participants at baseline according to vitamin D₃ and marine n-3 FAs assignments.

Characteristic	Total (<i>N</i> = 1031)	Vitamin D ₃			n-3 FAs		
		Vitamin D ₃ (<i>N</i> = 508)	Placebo (<i>N</i> = 523)		n-3 FAs (<i>N</i> = 514)	Placebo (<i>N</i> = 517)	
Age (y)	64.9 $\pm$ 6.3	64.7 $\pm$ 6.3	65.1 $\pm$ 6.5	0.290	64.9 $\pm$ 6.5	65.0 $\pm$ 6.4	0.841
Female sex – <i>N</i> (%)	506 (49.3)	252 (49.6)	254 (48.6)	0.738	253 (49.2)	253 (48.9)	0.927
Race – <i>N</i> (%)				0.698			0.432
White	854 (84.6)	420 (84.2)	434 (85.1)		420 (83.8)	434 (85.4)	
Black	83 (8.2)	40 (8.0)	43 (8.4)		40 (8.0)	43 (8.5)	
Others	72 (7.2)	39 (7.8)	33 (6.5)		41 (8.2)	31 (6.1)	
BMI – kg/m ²	28.2 $\pm$ 5.3	28.1 $\pm$ 5.3	28.3 $\pm$ 5.3	0.464	28.6 $\pm$ 5.4	27.8 $\pm$ 5.3	0.013
Current smoking – <i>N</i> (%)	55 (5.4)	30 (6)	25 (4.8)	0.408	26 (5.1)	29 (5.7)	0.667
Medication use – <i>N</i> (%)							
Hypertension	424 (41.1)	198 (39.0)	218 (43.2)	0.167	216 (42.0)	208 (40.2)	0.559
Diabetes	88 (8.5)	48 (9.5)	40 (7.7)	0.301	44 (8.6)	44 (8.5)	0.977
Cholesterol	360 (34.9)	178 (34.0)	173 (35.8)	0.546	192 (37.4)	168 (32.5)	0.102
25(OH)D – ng/mL	28.2 $\pm$ 9.0	27.6 $\pm$ 8.8	28.7 $\pm$ 9.2	0.056	28.3 $\pm$ 9.4	28.1 $\pm$ 8.6	0.660
25(OH)D < 20 ng/mL – %	861 (83.6)	425 (83.8)	436 (83.4)	0.842	425 (82.9)	436 (84.3)	0.519
n-3 index – %	3.0 $\pm$ 1.0	2.9 $\pm$ 1.0	3.0 $\pm$ 1.0	0.530	3.0 $\pm$ 1.0	2.9 $\pm$ 1.0	0.380
SBP (mmHg)	124.3 $\pm$ 14.4	123.9 $\pm$ 14.3	124.7 $\pm$ 14.5	0.401	124.4 $\pm$ 14.3	124.2 $\pm$ 14.6	0.827
DBP (mmHg)	76.2 $\pm$ 9.2	75.8 $\pm$ 9.1	76.6 $\pm$ 9.3	0.167	76.6 $\pm$ 9.0	75.8 $\pm$ 9.4	0.145
LTL – kb	8.7 $\pm$ 1.4	8.8 $\pm$ 1.4	8.7 $\pm$ 1.5	0.599	8.7 $\pm$ 1.4	8.7 $\pm$ 1.4	0.885

Statistics display as mean $\pm$ SD for continuous variables, and *N* (%) for categorical variables.

Abbreviations: CTSC, Clinical and Translational Science Center; 25(OH)D, 25-hydroxyvitamin D; n-3 FAs, omega-3 fatty acids; SBP, systolic blood pressure; DBP, diastolic blood pressure; LTL, leukocyte telomere length.

between the placebo group and treatment group were tested using two-tailed t-test for continuous variables with normal distribution and Pearson's chi-squared for categorical variables.

Missing data: We calculated the proportion of available and missing cases (due to loss of follow-up, no consent for genetic testing, and missing LTL values) at baseline, year 2, and year 4 and tested if the available data were well balanced between treatment and placebo groups by chi-squared test.

The primary outcome was the changes in LTL. We prespecified that we would analyze vitamin D₃ or marine n-3 FAs supplementation separately, as was done in the main VITAL trial. The effects of vitamin D₃ or marine n-3 FAs supplementation on changes in LTL were estimated by linear mixed-effect models for repeated measures adjusted for baseline covariates and other interventions as fixed covariates with an

interaction between randomized intervention arm and time (baseline, year 2, or year 4). We used Shapiro-Wilks and other visual tests to check the normality distribution of the residuals in the model. The normality assumption of residuals was met in our model. Autoregressive structure of order 1 of the within-group errors was assumed to account for successive observations with the groups. Three models were constructed to control potential confounding factor imbalance. Model 1 was adjusted for the randomization group for the other treatment (intention-to-treat); model 2 (Primary analysis) was adjusted for age, sex, race, and BMI, in addition to model 1; model 3 was adjusted for the covariates in model 2 and smoking, and taking medications for hypertension, diabetes, and high cholesterol. We further conducted a trend analysis by including an ordinal interaction term (treatment × time) in the fixed effects model.

TABLE 2

Effects of vitamin D₃ or marine n-3 FAs supplementation on leukocyte telomere length overtime.

Model 1 N = 1031				Placebo			Diff. in pre-post changes (95% CI)	P value
Time point	Mean (95% CI)	Pre-post change (95% CI)	P value	Mean (95% CI)	Pre-post change (95% CI)	P value		
Baseline	8.78 (8.65, 8.90)	—	—	8.73 (8.61, 8.86)	—	—	—	—
Year 2	8.73 (8.59, 8.86)	−0.05 (−0.14, 0.02)	0.153	8.61 (8.49, 8.74)	−0.12 (−0.20, −0.04)	0.005	0.07 (−0.05, 0.18)	0.255
Year 4	8.75 (8.62, 8.89)	−0.02 (−0.12, 0.07)	0.616	8.57 (8.43, 8.71)	−0.16 (−0.26, −0.07)	0.001	0.14 (0.01, 0.27)	0.041
Trend per year ¹							0.034 (0.002, 0.07)	0.039
Model 2 N = 1009								
Baseline	8.78 (8.65, 8.90)	—	—	8.75 (8.62, 8.87)	—	—	—	—
Year 2	8.73 (8.6, 8.86)	−0.05 (−0.13, 0.03)	0.174	8.62 (8.50, 8.75)	−0.12 (−0.20, −0.04)	0.004	0.07 (−0.04, 0.19)	0.228
Year 4	8.75 (8.62, 8.89)	−0.02 (0.05, 0.07)	0.628	8.58 (8.44, 8.72)	−0.16 (−0.26, −0.07)	0.001	0.14 (0.01, 0.27)	0.039
Trend per year ¹							0.035 (0.002, 0.07)	0.037
Model 3 N = 1002								
Baseline	8.78 (8.65, 8.91)	—	—	8.75 (8.62, 8.87)	—	—	—	—
Year 2	8.73 (8.60, 8.86)	−0.05 (−0.13, 0.04)	0.274	8.62 (8.49, 8.75)	−0.12 (−0.20, −0.04)	0.004	0.08 (−0.04, 0.19)	0.211
Year 4	8.76 (8.62, 8.90)	−0.02 (−0.11, 0.07)	0.697	8.58 (8.44, 8.71)	−0.17 (−0.26, −0.07)	0.001	0.15 (0.02, 0.28)	0.027
Trend per year ¹							0.04 (0.005, 0.07)	0.025
Model 1 N = 1031								
Time point	Mean (95% CI)	Pre-post change (95% CI)	P value	Mean (95% CI)	Pre-post change (95% CI)	P value	Diff. in pre-post changes (95% CI)	P value
Baseline	8.77 (8.64, 8.90)	—	—	8.74 (8.61, 8.87)	—	—	—	—
Year 2	8.69 (8.56, 8.82)	−0.08 (−0.16, 0.00)	0.054	8.65 (8.52, 8.78)	−0.09 (−0.17, −0.01)	0.037	0.01 (−0.11, 0.12)	0.924
Year 4	8.71 (8.57, 8.84)	−0.06 (−0.16, 0.03)	0.180	8.62 (8.48, 8.76)	−0.12 (−0.21, −0.02)	0.013	0.05 (−0.08, 0.18)	0.423
Trend per year ¹							0.01 (−0.02, 0.05)	0.451
Model 2 N = 1009								
Baseline	8.78 (8.65, 8.91)	—	—	8.74 (8.61, 8.87)	—	—	—	—
Year 2	8.70 (8.57, 8.83)	−0.08 (−0.17, 0.01)	0.052	8.65 (8.52, 8.78)	−0.09 (−0.17, −0.00)	0.039	0.00 (−0.11, 0.12)	0.953
Year 4	8.72 (8.58, 8.86)	−0.06 (−0.16, 0.03)	0.183	8.62 (8.48, 8.76)	−0.12 (−0.21, −0.02)	0.014	0.05 (−0.08, 0.19)	0.428
Trend per year ¹							0.01 (−0.02, 0.05)	0.459
Model 3 N = 1002								
Baseline	8.78 (8.65, 8.91)	—	—	8.74 (8.62, 8.87)	—	—	—	—
Year 2	8.70 (8.57, 8.83)	−0.08 (−0.16, 0.00)	0.060	8.66 (8.52, 8.79)	−0.09 (−0.17, −0.01)	0.035	0.01 (−0.11, 0.13)	0.883
Year 4	8.72 (8.58, 8.86)	−0.06 (−0.16, 0.03)	0.191	8.62 (8.49, 8.76)	−0.12 (−0.22, −0.03)	0.012	0.06 (−0.08, 0.19)	0.396
Trend per year ¹							0.01 (−0.02, 0.05)	0.421

Effect of vitamin D₃ or n-3 FAs supplementation on telomere length was estimated by the linear mixed model for repeated measures. Model 1 was adjusted for the randomization group for the other treatment; model 2 was adjusted for age, sex, race, and BMI in addition to model 1; model 3 was adjusted for the covariates in model 2, smoking status, and taking medications for hypertension, diabetes, or lowering cholesterol.

Abbreviation: CI, confidence interval.

¹ The overall trend was estimated by including an ordinal interaction term (treatment × time) in the fixed effects model.

In addition, we explored effect modification by the following baseline covariates: age group (<median of 64.4 y or ≥ 64.4 y), sex (male, female), race (White, Black or other), BMI (<30 kg/m² or ≥ 30 kg/m²), 25(OH)D (<20 ng/mL or ≥ 20 ng/mL), ω -3 index (<median of 2.7% or $\geq 2.7\%$), smoking status, medications for high cholesterol, diabetes, hypertension, and the other treatment arm (active or placebo). However, there was no control for multiple testing, and no formal adjustment was made to the *P* values or confidence interval. Thus, these subgroup analysis results should be considered exploratory and interpreted with caution. A *P* < 0.05 was considered statistically significant. All analyses were performed using Stata version 18.0 (StataCorp).

Results

General characteristics of the participants

A total of 2571 LTL measurements (993 from baseline, 918 from year 2, and 660 from year 4) from 1031 participants (aged 64.9 ± 6.5 y, 49% female, 84% White, and 9% Black) were included in this study (Figure 1). Table 1 presents the baseline characteristics according to vitamin D₃ and marine n-3 FAs assignments. There was no significant difference in the baseline general characteristics between the vitamin D₃ active group and vitamin D₃ placebo group or between the marine n-3 FAs active group and n-3 FAs placebo group (*P* > 0.05), except that BMI was higher in the n-3 FAs active group than that n-3 FAs placebo group (*P* = 0.007).

There were 5.8% missing at baseline, 12.9% missing at year 2, and 37.4% missing (due to loss to follow-up, absence of consent for genetic analysis, and missing LTL values, Figure 1). There was no significant difference in the available cases between vitamin D₃ or n-3 fatty acids treatment group and placebo group at baseline, year 2 or year 4 (all *P* values > 0.05, Supplementary Tables S1–S4).

Effects of supplementation of vitamin D₃ and marine n-3 fatty acids on changes in LTL in all participants

The average LTL at baseline was 8.7 ± 1.4 kb. Table 2 presents the effects of vitamin D₃ and marine n-3 FAs supplementation on LTL. In the placebo group, substantial LTL attrition was observed both at year 2 (−0.12 [95% CI: −0.20, −0.04], *P* = 0.004) and year 4 (−0.16 [−0.26, −0.07] *P* = 0.001), but this was minimal in the vitamin D group either at year 2 (−0.05 [−0.13, 0.03], *P* = 0.174) or at year 4 (−0.02 [0.05, 0.07], *P* = 0.628). Compared with the placebo group, vitamin D₃ supplementation significantly decreased LTL attrition by 0.14 kb (0.01, 0.27) over 4 years, *P* = 0.039 (Figure 2A). LTL was preserved 140 bp longer in vitamin D₃ group than in placebo group over 4-y period. Marine n-3 FAs supplementation did not demonstrate a significant difference in change of LTL compared with placebo at either year 2 or year 4 (*P* > 0.05, Figure 2B).

Overall trend analysis

The overall trend analysis showed that vitamin D₃ supplementation group had LTLs that were ~0.035 kb higher per year of follow-up than placebo group (*P* = 0.037, Table 2) adjusting for the same covariates of model 2, which equates to 140 bp over the 4-y period. All 3 models were statistically significant (all *P* < 0.05). The n-3 FAs supplementation group did not show a significant effect over time (*P* > 0.05, Table 2).

Exploratory subgroup analysis

Figure 3 shows the results of the exploratory subgroup analyses of the associations of vitamin D₃ supplementation and LTL at year 4.

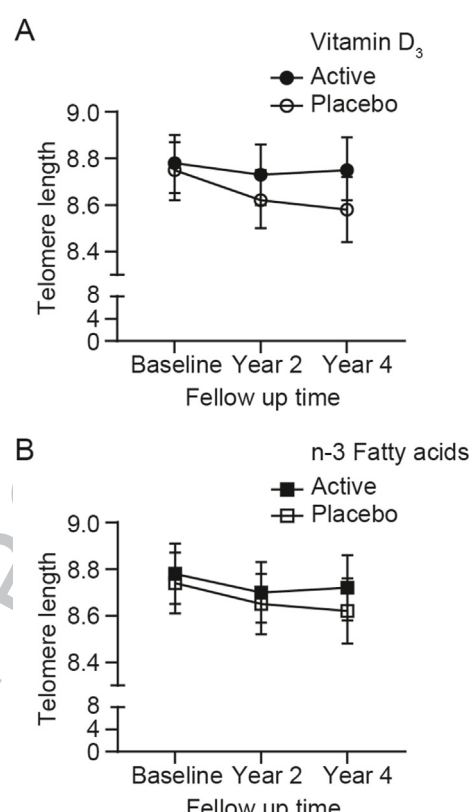


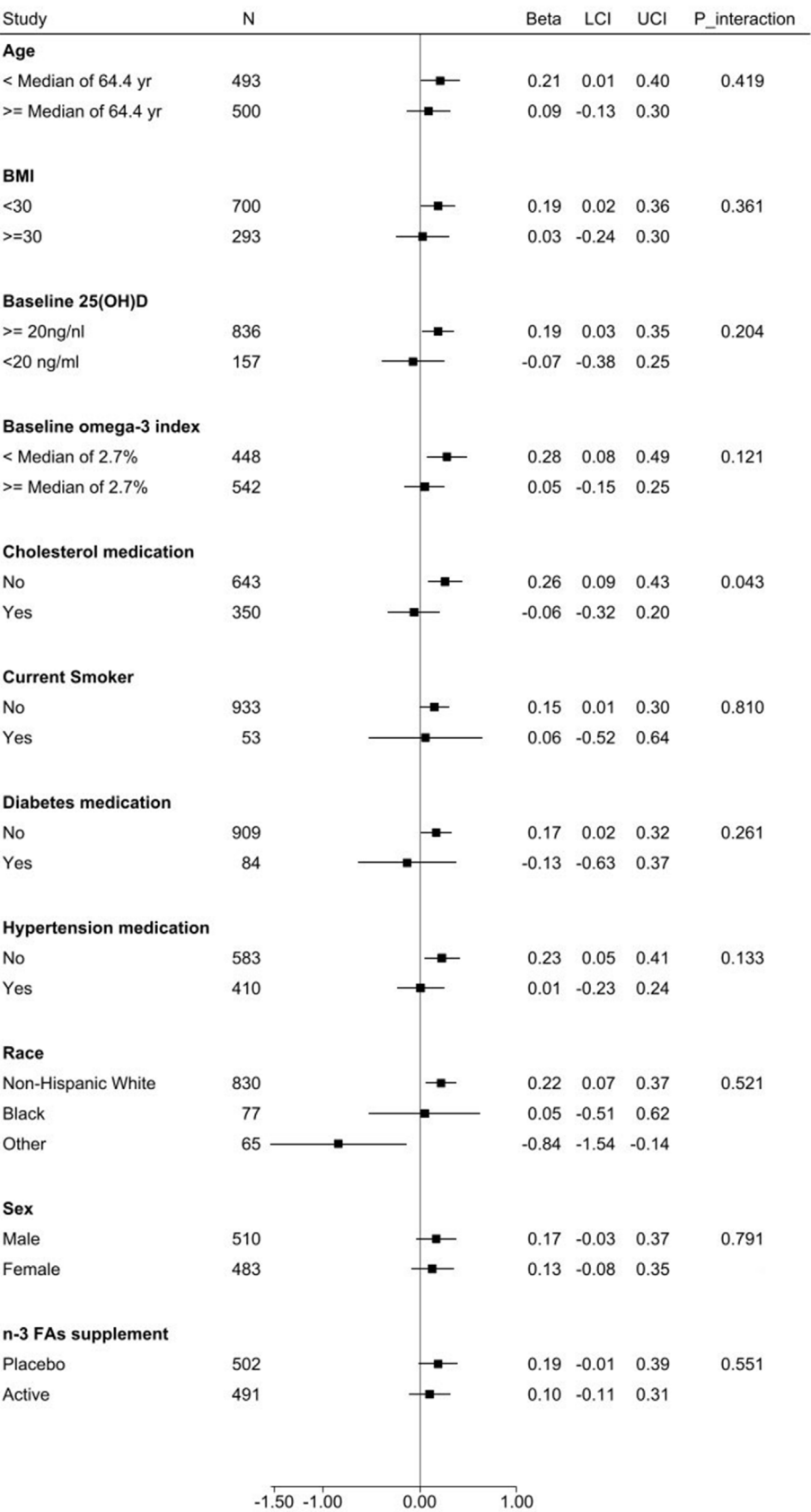
FIGURE 2. (A) Effect of vitamin D₃ supplementation on leukocyte telomere length overtime. The effect of vitamin D₃ on LTL was estimated by the linear mixed model for repeated measures. Estimated mean and 95% confidence interval from model 2 were presented. (B) Effect of marine n-3 FAs supplementation on leukocyte telomere length overtime. The effect of marine n-3 FAs on LTL was estimated by the linear mixed model for repeated measures. Estimated mean and 95% confidence interval from model 2 were presented.

Vitamin D₃ supplementation significantly decreased LTL attrition among the younger group (age < median of 64.4 y, *P* = 0.038), Whites (*P* = 0.004), other race (*P* = 0.019), nonsmokers (*P* = 0.002), non-obese (BMI < 30 kg/m², *P* = 0.03), lower baseline 25(OH)D (*P* = 0.03) and n-3 index (*P* = 0.005), and the participants that were not on medications for high cholesterol (*P* = 0.003), diabetes (*P* = 0.025), and hypertension (*P* = 0.012), although only the interaction with cholesterol medication and other race (compared with non-Hispanic White race) were statistically significant. There was no interaction between vitamin D₃ and n-3 FAs supplementation on LTL (*P* = 0.527, Figure 3).

Discussion

In the VITAL-CTSC subcohort, we found that vitamin D₃ supplementation significantly reduced telomere attrition over a 4-y period, preventing 140 bp of LTL loss compared with placebo. Overall trend analysis also showed that vitamin D₃ supplementation group had LTLs that were ~0.035 kb higher per year of follow-up compared with placebo group, which equates to 140 bp over the 4-y period. However, we did not observe any significant effect of n-3 FAs on LTL either at year 2 or year 4 or in overall trend analysis.

In the TwinUK cohort of 2160 British females aged 18–79 y (mean age: 49.4), Richards et al. [4] found that higher serum 25(OH)D



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concentrations were associated with longer LTL. Moreover, the difference in multiply adjusted LTL between the highest and lowest serum vitamin D tertiles was similar in magnitude to the difference in LTL associated with 5 y of chronologic age [4]. In 1424 females of the Nurses' Health Study, Liu et al. [3] showed a positive correlation between LTL and serum 25-OHD concentrations. Logistic regression analysis indicated a concentration-dependent relationship [3]. Data from the US NHANES 2001–2002 by Beilfuss et al. [5] showed that an increment in serum 25(OH)D of 10 nmol/L was associated with a 0.03 ± 0.01 -kbp longer LTL, adjusted for age, sex, race/ethnicity, and other factors ($P = 0.001$). In the same age group, 25(OH)D concentrations ≥ 50 nmol/L were associated with a 0.13 ± 0.04 -kbp longer LTL than those for 25(OH)D concentrations < 50 nmol/L ($P = 0.01$) (5). Further support for vitamin D supplementation preserving telomere length comes from 2 small randomized controlled trials. Borras et al. [23] observed longer telomeres in hemodialysis patients treated with calcitriol for ≥ 6 mo compared with hemodialysis patients without such treatment. In 183 Chinese patients aged ≥ 65 y with mild cognitive impairment, 12-mo vitamin D supplementation increased LTL and improved cognitive function [11].

Vitamin D₃ supplementation prevented 140 bp of LTL loss over 4-y period, which might be clinically significant. A previous study measured absolute LTL of 510 participants in a longitudinal cohort and found an average of 460 bp decrease in LTL over 10 y [24]. Therefore, 140 bp preserved by vitamin D₃ supplementation in our study could mean a 3-y decrease in aging.

Vitamin D and its receptors play an essential role in aging prevention due to its antiproliferative, anti-inflammatory, and apoptotic properties. However, the exact underlying mechanisms of preserving telomeres by vitamin D supplementation are not well understood. Nonetheless, there are several potential mechanistic pathways that are worth further investigation. The prevention of telomere shortening by vitamin D supplementation could be mediated by its increased telomerase activity [25]. Telomerase is an enzyme that elongates telomeres by adding TTAGGG sequence to the ends of existing chromosomes. We previously conducted a double-blind, randomized, and placebo-controlled clinical trial of 16-wk vitamin D supplementation (60,000 IU/mo [equivalent to ~ 2000 IU/d]) in overweight African Americans, peripheral blood mononuclear cell telomerase activity increased by 19.2% from baseline (1.56 ± 0.29 AU) to posttest (1.86 ± 0.42 AU, $P < 0.0001$) in the vitamin D-treated group compared with the placebo group. [10] Vitamin D receptor is a member of the nuclear receptors superfamily of ligand-activated transcription factors, which is viewed as a master regulator of transcription. Vitamin D possibly improves cellular aging via a variety of physiological pathways, including NF- κ B inhibition [26,27] and PI3K/Akt activation [28]. Telomerase has become a downstream target in the PI3K/Akt pathway [29,30].

Telomeres regions are favored targets for random DNA damage [31]. Telomeres are particularly sensitive to oxidative stress compared with the bulk of the genome [32] and less efficiently repaired when subjected to single-stranded breaks and UV-induced damage [33]. Vitamin D deficiency has been associated with elevated DNA damage [34,35], particularly in patients with hyperglycemia and cancer [36]. A recent systematic review of both human and animal studies reported a strong association between adequate vitamin D concentrations and

reduced DNA damage, suggesting that sufficient vitamin D intake or supplementation may help protect against DNA damage [37]. It is possible that vitamin D supplementation preserves telomere length observed in the current study through protection against DNA damage.

LTL has been proposed as a potential biomarker of premature aging of the immune system, known as immunosenescence, which underlies the development of autoimmune diseases, cancer, and several other chronic diseases. Indeed, among patients with some autoimmune disease, loss of normal immune regulation and "premature immune senescence" have been described [38]. Telomere erosion appears to proceed more rapidly in subjects with autoimmune diseases [39]. Emerging evidence suggests that telomeres are new players in immune-mediated inflammatory diseases [40]. A meta-analysis demonstrated that telomere length was significantly shorter in patients with SLE, regardless of ethnicity, sample type, or assay method evaluated [41]. Two meta-analyses confirmed the presence of short TL in patients diagnosed with rheumatoid arthritis [42,43]. A recent review reported that shortened telomere length is associated with several autoimmune diseases, including inflammatory bowel disease, psoriasis, rheumatoid arthritis, spondylarthritis and uveitis [40]. If telomere shortening occurs years prior to the onset of clinical autoimmune disease, there might be a window for prevention and early intervention.

The recent VITAL ancillary study (NTC01351805) by Hahn et al. [16] demonstrated that vitamin D supplementation for 5 y with or without marine n-3 FAs reduced all incident autoimmune diseases by 22% with more pronounced effects found after a latency of 2 y and among those with lower BMI, whereas n-3 FAs supplementation with or without vitamin D reduced the autoimmune disease rate by 15% (not statistically significant). Moreover, we previously reported that vitamin D₃ supplementation with or without n-3 FA supplementation decreased circulating level high sensitivity C-reactive protein by 19% at year 2 in VITAL-CTSC participants [17]. The beneficial effect of vitamin D supplementation on telomere length, as observed in the present study, may mediate the effect of vitamin D supplementation on autoimmune disease.

In the subgroup analysis, we found a significant interaction between cholesterol medication and vitamin D₃ supplementation on LTL. Vitamin D₃ supplementation decreased LTL shortening rate only among the ones not on cholesterol medication. The participants on cholesterol medication could have abnormal cholesterol, and a higher cholesterol concentration is associated with a lower 25(OH)D concentration [44]. In addition, the significant different impact of vitamin D₃ supplementation on LTL could be caused by the cholesterol medication-induced changes in circulating vitamin D response to vitamin D₃ supplementation. A clinical trial showed that statin, one of the major lipid-lowering medications, moderated the circulating vitamin D response to vitamin D₃ supplementation [45]. In their study, statin users had a 21.4% smaller increase in 25(OH)D serum concentrations than nonusers after 12 mo of vitamin D₃ supplementation among 646 older adults [45]. We previously found that the baseline methylation level of cg07873128 (*OSBPL5*) that regulates cholesterol balance was also significantly associated with serum 25(OH)D response to 16 wk vitamin D₃ supplementation among African Americans [46].

FIGURE 3. Effect of vitamin D₃ on 4-y changes and 95% confidence intervals of leukocyte telomere length by baseline subgroups and n-3 FAs intervention. Estimates were from linear mixed model. Models were adjusted for n-3 fatty acid randomization group (intention-to-treat analysis), age, sex, race, BMI, smoking, and taking medications for hypertension, diabetes, and lowering high cholesterol. Analyses were not adjusted for multiple comparisons. 25(OH)D, 25-hydroxyvitamin D; n-3 FAs, omega-3 fatty acids; LCI, lower bound of 95% confidence interval; UCI, upper bound of 95% confidence interval.

Although no significant interaction with BMI was found, vitamin D₃ supplementation significantly decreased LTL attrition in the non-obese group. Interestingly, in the initial VITAL publication, 2000 IU/d vitamin D₃ supplementation was associated with lower incidence of total invasive cancer among normal weight individuals (HR: 0.76; 95% CI: 0.630-90) [19], but not those at higher weights. Chandler et al. [15] also reported that vitamin D supplementation significantly reduced the incidence of metastatic/fatal cancer in the normal BMI group. Both studies found BMI was a significant modifier for total cancer and advanced cancer in VITAL [15,19]. Furthermore, vitamin D supplementation for 5 y with or without marine n-3 FAs reduced all incident autoimmune diseases by 22%, with more pronounced effects found after a latency of 2 y and among those with lower BMI [16]. Adiposity is a major repository and has large capacity for fat sequestration of vitamin D in the body [47], which could reduce bioactivity of vitamin D in the circulation to dilute the effect of vitamin D treatment on LTL. Additionally, meta-analyses show that obesity could accelerate LTL shortening [48,49]. The strength of our study includes that we leveraged the large randomized, double-blind, placebo-controlled VITAL trial. Moreover, the effects of vitamin D₃ and marine n-3 FAs supplementations on LTL were evaluated for 2–4 y, a longer duration of treatment than in most previous trials, showing a dynamic response to the supplementation longitudinally. We used biospecimen collected from the VITAL-CTSC participants, who had inperson visits and blood samples were drawn on-site and promptly processed. The present study also has several limitations. First, majority participants were non-Hispanic whites, therefore, the findings may not be generalizable to other racial/ethnic groups. Second, it was a post hoc analysis of a completed randomized controlled trial, which was not originally designed to test the effect of vitamin D₃ and marine n-3 FAs supplementations on LTL. Third, although 23 participants with no LTL data at all were excluded from the analysis, the remaining 1031 participants were still evenly distributed between treatment and placebo groups. Fourth, we did not detect a significant beneficial effect of marine n-3 FAs on LTL, which could be due to reduced power resulted from 37% missing cases at year 4 follow-up. Last, results from subgroup analyses had limited power, were exploratory, and should be considered hypothesis-generating and interpreted with caution given that they are not adjusted for multiple comparison.

To conclude, in the VITAL Telomere study, vitamin D₃ supplementation reduced telomere attrition and preserved telomere length, supporting an anticellular aging effect of vitamin D. The effect of n-3 FAs supplementation on LTL was not significant over a 4-y period. Our findings underscore the potential benefits of vitamin D on biological aging and age-related diseases. Future independent large-scale RCTs are warranted to replicate our findings, particularly in other racial/ethnic groups.

Author contribution

The authors' responsibilities were as follows – YD, HZ, NC, JM: designed research; YH and HZ: conducted research; WC, JM: provided essential reagents or provided essential materials; YH, HZ: performed LTL analysis; LC, KK, NC, BL, WL: analyzed data or performed statistical analysis; YD, LC, HZ: drafted paper; JM, NC, WC, BL, IML: revised paper. HZ, YD, JM are responsible for design, writing and final content; and all authors: have read and approved the final version of the manuscript.

Conflict of interest

The authors report no conflicts of interest.

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Data availability

Data described in the manuscript, code book, and analytic code will be made available upon request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ajcnut.2025.05.003>.

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